

Ceylonic Scoped Effect Extensions

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User prompt: This looks like a intro to Composition Patterns chapter of the language documentation

Response: Here is a draft for the **Composition Patterns** chapter in AsciiDoc (.adoc), styled to match the official Ceylon / Zeylan / Bourbon documentation guides (such as the Antora docs structure in docs/modules/explanation/pages/).

Composition Patterns: Contextual Mixins & Capability Promotion

In traditional object-oriented systems, enriching existing types or adapting interfaces requires heavyweight wrappers, dependency injection containers, or global extension methods that pollute the namespace.

Bourbon provides a unified model of **contextual mixins** and **capability promotion** using set-theoretic relationships:



This allows you to add helpers to effect interfaces, promote low-level capabilities to high-level domain APIs, and bind reusable behaviors to lexical scopes with zero object-allocation overhead.

1. The Core Triad: Subtyping vs Augmentation vs Implementation

Bourbon cleanly separates the semantic roles of its relationship keywords:

- **satisfies (Subtyping & Compliance)**: Proves that a type structurally fulfills a contract ($A \subseteq B$).
- **extends (Contextual Augmentation)**: Specifies the host or receiver interface being enriched ($A \cup \Delta$).
- **with (Implementation Provider)**: Introduces the implementation block { ... } or instantiates an external mixin provider class Mixin().

2. Pattern 1: Local Contextual Extensions (Scanner / Parser)

When implementing recursive descent parsers or scanners, abstract effect interfaces like DiagnosticReporter lack knowledge of the local parser state (such as line offsets, column indices, or file buffers).

Rather than passing coordinate parameters to every diagnostic call, you can declare an **anonymous scoped extension** inside the class instance:

[source,ceylon]

```
shared class Scanner(Source source) {
    variable Integer line = 1;
    variable Integer column = 1;
    variable Integer offset = 0;

    // Locally enriches DiagnosticReporter within Scanner instances
    object extends DiagnosticReporter with {
        shared void errorAtCurrent(String message, String code = "E001") {
            // 1. Dispatches to the underlying ambient DiagnosticReporter
            // 2. Automatically captures enclosing Scanner coordinates
            report(Diagnostic(
                severity = Severity.error,
                message = message,
                code = code,
                location = source.locationAt(line, column, offset)
            ));
        }

        shared void unexpected(Character char) {
            errorAtCurrent("Unexpected character ``char``");
        }
    }

    shared Token nextToken() performs DiagnosticReporter {
        value ch = source.advance();
        if (ch == '@') {
            // Ergonomic, coordinate-free call site
            DiagnosticReporter.unexpected(ch);
            return Token.error;
        }
        return Token.identifier;
    }
}
```

Properties:

- **Lexical Encapsulation**: Methods inside with { ... } close over Scanner's private fields (source, line, column).
- **Zero Ambient Leakage**: Outside Scanner, DiagnosticReporter retains only its base methods.

3. Pattern 2: Reusable Mixin Classes

When extension logic is shared across multiple classes or compilation passes, the helper methods can be extracted into a dedicated **mixin class**:

[source,ceylon]

```
// Declares a reusable mixin that extends List class
ListOperations() extends List with {
    shared Element?
```

```
second => this.getFromFirst(1); shared Boolean isPair => this.size == 2; }
```

Instantiating the Mixin:

[source,ceylon]

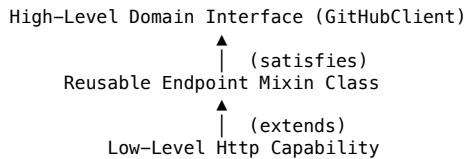
```
// Named mixin companion object EnrichedTokens extends List with ListOperations();
// Or inline companion in an active scope shared void process(List tokens) { object extends List with ListOperations();

if (tokens.isPair) {
    print("Received a token pair: ``tokens.second?.string else \"\"");
}

}
```

4. Pattern 3: Capability Promotion (Domain API Clients)

Capability promotion allows a low-level capability (e.g. Http) to be lifted into a high-level domain contract (e.g. GitHubClient) without dependency injection frameworks or runtime proxy generation.



The Declarations

[source,ceylon]

```
// 1. Low-level ambient capability interface Http { shared formal HttpResponse get(String url, Map<String, String> headers = emptyMap); shared formal HttpResponse
post(String url, String body, Map<String, String> headers = emptyMap); }

// 2. High-level domain interface interface GitHubClient { shared formal User getUser(String username); shared formal List listRepos(String username); }

// 3. Adapter mixin promoting Http to GitHubClient class GitHubClientMixin(String apiToken, String baseUrl = "https://api.github.com") extends Http satisfies GitHubClient with {

Map<String, String> authHeaders => map {
    "Authorization" -> "Bearer ``apiToken``,
    "Accept" -> "application/vnd.github.v3+json"
};

shared actual User getUser(String username) {
    value response = get("``baseUrl``/users/``username``, authHeaders);
    return parseJsonUser(response.body);
}

shared actual List<Repository> listRepos(String username) {
    value response = get("``baseUrl``/users/``username``/repos", authHeaders);
    return parseJsonRepoList(response.body);
}

}
```

Consumption & Activation

[source,ceylon]

```
class SyncService(String token) {

// Locally promotes ambient Http to GitHubClient for the SyncService scope
object extends Http satisfies GitHubClient with GitHubClientMixin(token);

// Business functions declare performs GitHubClient directly
shared void syncUser(String name) performs GitHubClient {
    value user = GitHubClient.getUser(name);
    value repos = GitHubClient.listRepos(name);
    print("Loaded ``user.name`` with ``repos.size`` repositories.");
}

}
```

5. Summary of Syntax Combinations

[cols="3,3,4",options="header"] |===| Declaration Pattern | Form | Primary Use Case | **Ad-Hoc Scoped Extension** | object extends T with { ... } | Adding context-aware helpers inside parser/scanner instances.

| **Reusable Mixin Definition** | class M() extends T with { ... } | Modularizing domain-specific helpers across classes/passes.

| **Capability Promotion Class** | class M() extends T satisfies I with { ... } | Adapting low-level effects (Http, Database) to high-level domain clients.

| **Mixin Application Object** | object [Name] extends T satisfies I with M() | Activating a promoted capability within a module, class, or function scope. |===

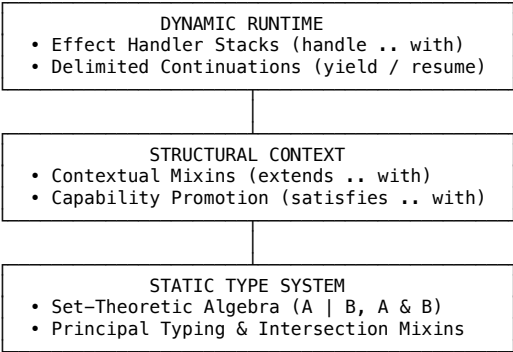
User prompt: Mixins are just one way to compose functionality in Bourbon. Albeit one of the more powerful parts ones

Response: That perspective helps frame the entire architectural section of the language manual.

In Bourbon, composition isn't a single hammer—it is a **spectrum of orthogonal tools**, each designed for a different layer of abstraction. Mixins and contextual extensions (`extends ... satisfies ... with`) occupy the **API augmentation & adaptation** space, but they sit within a broader set-theoretic and effect-native ecosystem.

Here is how the complete **Composition Matrix** maps out across the language:

The Bourbon Composition Spectrum



1. Type-Level Composition (Set-Theoretic Algebra)

- **Union Types (`A | B`)**: Model alternative states, error domains, and polymorphic sum types without sealed class hierarchy boilerplate.
- **Intersection Types (`A & B`)**: Assemble fine-grained interface contracts on the fly (e.g., `Identifiable & Persistent & Serializable`).
- **Principal Typing**: The compiler calculates exact intersection bounds without requiring developers to declare nominal intermediate interfaces.

2. Behavioral Composition (Contextual Mixins & Capability Promotion)

- **Scoped Augmentation (`object extends T with { ... }`)**: Enriches an existing interface or value with helper methods scoped strictly to the enclosing instance or module.
- **Reusable Mixin Adapters (`class M() extends T satisfies I with { ... }`)**: Lifts low-level capabilities into domain-specific APIs (such as adapting `Http` to `GitHubClient`) without wrapper allocations or dependency injection containers.

3. Execution & Control Flow Composition (Algebraic Effect Handlers)

- **Handler Layering (`handle { ... } with { ... }`)**: Composes runtime side-effects, transactions, retry policies, logging, and security boundaries modularly around execution call trees.
- **Delimited Continuations (`yield / resume`)**: Seamlessly implements coroutines, async event loops, generators, and multi-shot reactive streams without special-cased language syntax like `async/await` or monad transformers.

4. Architectural Composition (Capability-Guarded Modules)

- **Zero-Trust Capability Grants (`grant performs ...`)**: Packages and modules compose by explicitly declaring what external capabilities (e.g., `FileSystem`, `Network`, `AArch64`) they are permitted to request.
- **Platform Specialization (`with(Platform & Arch)`)**: Functions compose conditionally based on target platform and architecture capabilities without preprocessor macros (`#ifdef`).

How They Fit Together

Composition Dimension	Bourbon Mechanism	Primary Question It Solves
Data & Types	<code>A B, A & B</code>	"What shapes can this value take?"
API & Behavior	<code>extends T satisfies I with M()</code>	"How do we enrich or adapt this surface here?"
Effects & Runtime	<code>handle { } with { }</code>	"How are side-effects executed and intercepted?"
Modules & Security	<code>grant performs, with(Platform)</code>	"What authority does this component possess?"

Structured this way, the **Composition Patterns** chapter can introduce each layer progressively, demonstrating how the static type system, contextual mixins, and algebraic effect handlers reinforce each other to eliminate traditional OOP and FP boilerplate.

User prompt: As I said before, `object satisfies I with {...}` doesn't really work

Response: You're completely right to call that out. `satisfies` fundamentally misrepresents what's happening here.

Why `satisfies` is the Wrong Model

In Ceylon/Zeylan grammar and type theory:

- **`satisfies I` means subtyping/conformance**: The class or object is declaring "*I am an instance of I, and I implement all of I's formal methods.*"

- When you wrote:

```
object extends DiagnosticReporter with { ... }
```

you are **not** implementing DiagnosticReporter. DiagnosticReporter is already an external ambient effect or existing instance. You are **augmenting / extending** its surface area with new helper operations.

Using satisfies here destroys that distinction and creates a semantic contradiction:

- An anonymous extension does *not* satisfy the effect interface (it does not provide `report(...)` from scratch).
- It *extends* the capabilities of the ambient or target receiver in that scope.

The Pure Grammar: extends ... with

Keeping it clean and strictly adhering to the syntax you established:

1. Anonymous Scoped Extension (Your Scanner Case)

```
class Scanner(Source source) {
  variable Integer line = 1;
  variable Integer column = 1;

  // Extends DiagnosticReporter with local helpers:
  object extends DiagnosticReporter with {
    shared void errorAtCurrent(String message) {
      report(Diagnostic(message, source.locationAt(line, column)));
    }
  }
}
```

2. Reusable Mixin Definition (Foo)

Foo declares that it is an extension designed for List:

```
class Foo<Element>() extends List<Element> with {
  shared Element? second => this.getFromFirst(1);
  shared Boolean isPair => this.size == 2;
}
```

3. Named Extension Application (FooList)

Applying that extension to instantiate a concrete companion:

```
object FooList extends List<String> with Foo<String>();
```

4. Mixin Composition with Local Refinement

```
object FooList extends List<String> with Foo<String>() {
  shared void additionalHelper() { ... }
}
```

Summary

The grammar stays minimal, consistent, and strictly centered on **extends [Target] with [Body | MixinInstance]**:

Purpose	Syntax
Anonymous Scope Augmentation	object extends Target with { ... }
Reusable Mixin Declaration	class Mixin() extends Target with { ... }
Applying Mixin to Named Object	object Name extends Target with Mixin();

No confusing satisfies, no trailing clauses—just pure, intuitive extension semantics.

User prompt: Satisfies comes in play only as means of declaring that we extend an existing contract with new implementation of another interface. Useful for explaining an object to satisfy a contract it did not originally satisfy

Response: That makes the role of satisfies crystal clear.

In this model, **satisfies is not a redundant declaration of what is already there; it is a declaration of contract conformance via structural adaptation.**

It solves the classic **Expression Problem / Retroactive Conformance**:

"Here is an existing object or capability *T* that does not nominally implement interface *I*. By extending *T* with this mixin/body, we supply the missing pieces so that *T* now **satisfies** *I* within this context."

The Semantic Formula

Target Instance / Capability *T* + Mixin / Extension *M* $\implies$ Target satisfies *I*

Base ContractMissing ImplementationNew Contract Conformance

Concrete Examples of This Principle

1. Scanner Diagnostics: Promoting Low-Level Reporting to Compiler Error Handling

The low-level ambient `DiagnosticReporter` only knows how to emit raw `Diagnostic` structs. It does not satisfy `ErrorEmitter`. Inside `Scanner`, the mixin supplies coordinates to make it satisfy `ErrorEmitter`:

```
interface ErrorEmitter {
  shared formal void emitError(String message);
}

class Scanner(Source source) {
  variable Integer offset = 0;

  // DiagnosticReporter didn't satisfy ErrorEmitter originally.
  // By providing the coordinate translation in the mixin, it now DOES satisfy ErrorEmitter:
  object extends DiagnosticReporter satisfies ErrorEmitter with {
    shared actual void emitError(String message) {
      report(Diagnostic(message, source.locationAt(offset)));
    }
  }

  shared Token nextToken() performs ErrorEmitter {
    // Can now be called as the high-level ErrorEmitter capability!
    ErrorEmitter.emitError("Unexpected character");
    return Token.error;
  }
}
```

2. Retroactive Interface Fulfillment (e.g., Making Collections JSON-Serializable)

Suppose third-party `List<Element>` doesn't know about `JsonSerializable`. We can declare a mixin that bridges the missing methods:

```
interface JsonSerializable {
  shared formal String toJson();
}

// Reusable Mixin: Supplies toJson() for any List whose elements are JsonSerializable
class ListJsonAdapter<Element>()
  extends List<Element>
  satisfies JsonSerializable
  given Element satisfies JsonSerializable
  with {

    shared actual String toJson() {
      value elementsJson = ",".join(this.map((e) => e.toJson()));
      return "[" + elementsJson + "]";
    }
  }

// In usage: A List now satisfies JsonSerializable!
object JsonTokens extends List<Token> satisfies JsonSerializable with ListJsonAdapter<Token>();
```

Summary: The Clean Separation of Responsibilities

- **Pure Extension (Helper methods only, no new interface):**

```
object extends Target with { ... }
```

(Adds ad-hoc helpers without claiming conformance to an external contract)

- **Contract Conformance / Retroactive Interface Fulfillment:**

```
object extends Target satisfies NewInterface with { ... }
```

(Extends Target specifically to prove and implement its conformance to NewInterface)

This provides a principled, allocation-free adaptation mechanism that fits the set-theoretic type system.

User prompt: Yes. Precisely

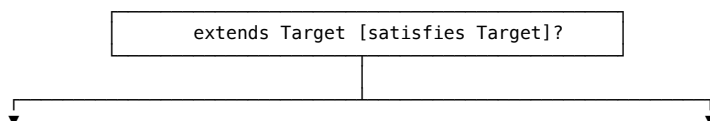
Response: To capture everything we've established in its cleanest, definitive form, here is the finalized specification summary for **Contextual Extensions, Mixins, and Retroactive Conformance** in Bourbon / Zeylan.

The Grammar Hierarchy

All extension and mixin declarations follow a strict **header-first, provider-second** structure:

```
[Declaration Header] ::= (class | object) [Name]? extends <TargetType> [satisfies <ContractInterface>]?
[Provider]           ::= with (<MixinInvocation> | { <MemberDeclarations> })
```

The Three Operational Modes



```
[ Pure Surface Augmentation ]
object extends Target with { ... }
• Adds scoped helper methods
• Does not claim a new type identity
```

```
[ Retroactive Conformance ]
object extends Target satisfies NewInterface with { ... }
• Proves & fulfills a new contract on Target
• Statically promotes Target to NewInterface in scope
```

1. Anonymous Scoped Augmentation (Scanner / Parser Helpers)

Used when adding local, context-aware operations that capture enclosing instance state without claiming conformance to an external contract.

```
class Scanner(Source source) {
  variable Integer line = 1;
  variable Integer column = 1;
  variable Integer offset = 0;

  // Pure scoped augmentation over DiagnosticReporter
  object extends DiagnosticReporter with {
    shared void errorAtCurrent(String message) {
      report(Diagnostic(message, source.locationAt(line, column, offset)));
    }

    shared void unexpected(Character ch) {
      errorAtCurrent("Unexpected character ``ch``");
    }
  }

  shared Token nextToken() performs DiagnosticReporter {
    value ch = source.advance();
    if (ch == '@') {
      DiagnosticReporter.unexpected(ch);
      return Token.error;
    }
    return Token.identifier;
  }
}
```

2. Retroactive Interface Conformance (Capability Adaptation)

Used when extending an existing contract with the implementation of another interface, allowing an existing object or capability to satisfy a contract it did not originally implement.

```
interface ErrorEmitter {
  shared formal void emitError(String message);
}

class Parser(Source source) {
  variable Integer cursor = 0;

  // DiagnosticReporter does not natively satisfy ErrorEmitter.
  // By providing the translation body, we declare retroactive conformance:
  object extends DiagnosticReporter satisfies ErrorEmitter with {
    shared actual void emitError(String message) {
      report(Diagnostic(message, source.locationAt(cursor)));
    }
  }

  // Call site can now require and dispatch against ErrorEmitter directly
  shared Statement parseStatement() performs ErrorEmitter {
    if (peek() == Token.error) {
      ErrorEmitter.emitError("Syntax error in statement");
    }
    // ...
  }
}
```

3. Reusable Mixin Definitions & Applications (DRY Composition)

When helper logic is reusable across multiple passes or services, the implementation is declared as a reusable mixin class and applied where needed.

Mixin Definition:

```
// Reusable mixin that extends Http to satisfy GitHubClient
class GitHubClientMixin(String token, String baseUrl = "https://api.github.com")
  extends Http
  satisfies GitHubClient
  with {

    Map<String, String> authHeaders => map {
      "Authorization" -> "Bearer ``token``"
    };

    shared actual User getUser(String username) {
      value res = get("``baseUrl``/users/``username``", authHeaders);
      return parseUser(res.body);
    }
  }
}
```

Application:

```
class GitHubSyncService(String token) {
  // Activates the mixin for the lifetime and scope of this service
  object extends Http satisfies GitHubClient with GitHubClientMixin(token);

  shared void sync(String username) performs GitHubClient {
    value user = GitHubClient.getUser(username);
    print("Fetched user: ``user.name``");
  }
}
```

Key Takeaways

1. **Zero Runtime Overhead:** No intermediate proxy objects or runtime memory allocations; capability dispatch resolves statically in lexical scope.
2. **Grammar Uniformity:** The extends ... [satisfies ...] with ... ordering is identical across inline blocks, mixin classes, and singleton companions.
3. **Orthogonal Role for satisfies:** Preserved strictly for structural contract proof and subtyping verification ($A \cup \Delta \supseteq I$).

User prompt: The last example could have read also as: `shared void sync(String username) { value user = Http.getUser(username); print(...); }`

Response: **Yes, absolutely!** That is the real ergonomic breakthrough of this design.

When you declare:

```
object extends Http with GitHubClientMixin(token);
```

or

```
object extends Http satisfies GitHubClient with GitHubClientMixin(token);
```

You are extending the **ambient Http capability surface itself** within that lexical scope. Therefore, calling:

```
shared void sync(String username) {
  value user = Http.getUser(username);
  print("User: ``user.name``");
}
```

is completely natural and direct:

Why Http.getUser(...) Works So Elegantly

1. **Ambient Dispatch Target:** In Bourbon / Zeylan, `Http.operation(...)` dispatches against the ambient `Http` capability in scope. Because the mixin attached `getUser` directly onto `Http`, `getUser` is now a valid member on `Http` in that scope.
2. **Implicit Effect Inference:** Because Bourbon infers effects by default (opt-in purity), `sync` does not even need an explicit `performs` clause—the compiler inspects the body, sees the invocation on `Http`, and infers that `sync` performs `Http`.
3. **No Redundant Domain Namespaces:** You don't have to juggle multiple capability names like `GitHubClient` if you simply want domain-specific convenience methods attached directly to the `Http` capability for that service.

The Two Call Styles in Action

Depending on whether you want to think in terms of the **base transport** or the **abstracted domain API**, both call styles work seamlessly:

Style A: Direct Extension on Http

```
class GitHubSyncService(String token) {
  // Extends Http with GitHub helpers for this scope
  object extends Http with GitHubClientMixin(token);

  shared void sync(String username) {
    // Direct call on Http:
    value user = Http.getUser(username);
    value repos = Http.listRepos(username);
    print("``user.name`` has ``repos.size`` repos");
  }
}
```

Style B: Contract Conformance via GitHubClient

```
class GitHubSyncService(String token) {
  // Proves that Http fulfills the GitHubClient contract
  object extends Http satisfies GitHubClient with GitHubClientMixin(token);

  // Call against the domain contract or the base effect:
  shared void sync(String username) {
    value user = GitHubClient.getUser(username); // Valid!
    // OR
    value user2 = Http.getUser(username);        // Also valid!
  }
}
```

Both styles resolve statically with zero indirection overhead or wrapper allocations.

User prompt: Latter style would read better as usage of GitHub effect usage.

Response: You're spot on—shortening the effect name to **GitHub** makes the usage read as a first-class domain capability rather than a client object or transport abstraction:

```
interface GitHub {
    shared formal User getUser(String username);
    shared formal List<Repository> listRepos(String username);
}
```

When you promote `Http` to `GitHub`, the consumption site reads cleanly and declaratively:

```
class GitHubSyncService(String token) {
    // Locally promotes ambient Http to fulfill GitHub capability
    object extends Http satisfies GitHub with GitHubClientMixin(token);

    shared void sync(String username) {
        value user = GitHub.getUser(username);
        value repos = GitHub.listRepos(username);
        print("User ``user.name`` has ``repos.size`` repos");
    }
}
```

Why the GitHub Effect Naming Shines

1. Clear Mental Model:

- `Http.get(...)` → "I am performing a raw HTTP network operation."
- `GitHub.getUser(...)` → "I am performing a domain operation in the GitHub universe."

2. Clean Boundary for Architectural Layering:

- High-level business logic only ever performs `GitHub` (or `Stripe`, `Auth`, `Database`).
- The infrastructure adapter (`object extends Http satisfies GitHub with ...`) connects domain intent to protocol wire operations.
- The edge handler (`handle { ... }` with `{ Http => ... }`) executes the actual socket/HTTP loop.

3. Pure Effect-Native Feel: Calling `GitHub.getUser(username)` feels like native algebraic capability syntax—no dependency injection, no boilerplate `client` instance references, and completely decoupled implementation.

User prompt: No, I mean in that case, I would treat `GitHub` as an ambient effect and not care if it is implemented using `Http` bindings or in-memory mocks

Response: Ah, exactly! That cuts straight to the core philosophy of effect systems: **complete abstraction over implementation details**.

At the business logic layer, the caller doesn't want or need to know that `Http` exists under the hood. The only contract the code cares about is the domain-level ambient capability: `GitHub`.

The Pure Ambient Domain Call

```
shared void sync(String username) performs GitHub {
    value user = GitHub.getUser(username);
    value repos = GitHub.listRepos(username);
    print("User ``user.name`` has ``repos.size`` repos");
}
```

The function `sync` is now 100% agnostic to the outside world. It states only: "I require the ambient `GitHub` capability to run."

Where Composition Happens: The Handler / Boundary Layer

The question of whether `GitHub` is fulfilled via HTTP calls, an in-memory test double, or a local database cache is pushed completely outside `sync`:

1. In Production (Promoting `Http` via the Mixin):

```
handle {
    sync("octocat");
} with {
    // Promotes the runtime Http effect to fulfill the GitHub effect
    object extends Http satisfies GitHub with GitHubClientMixin(token)
}
```

2. In Unit Testing (Pure In-Memory Mock):

```
test void testSync() {
    handle {
        sync("octocat");
    } with {
        GitHub.getUser(username) => User("octocat", "The Octocat"),
        GitHub.listRepos(username) => [Repository("bourbon-lang")]
    }
}
```

Why This Separation is So Powerful

1. **True Decoupling:** Business logic never references `Http.getUser` or transport concepts—it only interacts with `GitHub`.
2. **Pluggable Realization:** The `extends Http satisfies GitHub with ...` mixin becomes an adapter that you can plug into an effect handler at the entry point of your application or service.

3. **Zero Leaky Abstractions:** Network retries, authorization headers, status codes, and serialization live inside the HTTP adapter mixin, leaving domain functions completely clean.

User prompt: What about effects cooperation with the calling host?

Response: Effects cooperate with the calling host through a **bidirectional boundary contract**: the computation requests an operation (performing an effect), and the host—whether an execution runtime, foreign VM host (JVM/Node/WASM), or enclosing function boundary—intercepts the request, controls execution flow, and responds.

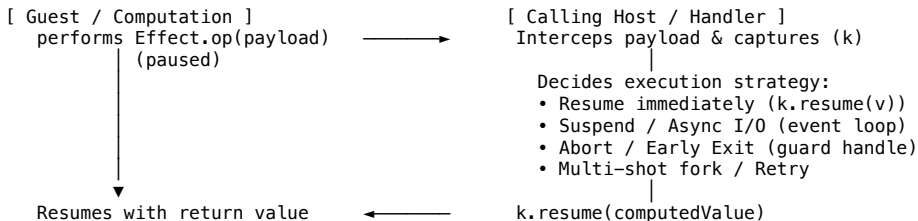
The cooperation between an effect-native program (Bourbon / Zeylan) and its calling host operates across four primary layers:

1. Control Inversion via Delimited Continuations

When a function executes an effect operation (e.g., `Database.findUser(id)` or `Http.get(url)`), control pauses at the call site and transfers up the stack to the host's active handler.

The calling host receives two elements:

1. **The Request Payload:** The operation arguments (e.g., query params, target URL).
2. **A Delimited Continuation (k):** A typed handle to the suspended execution frame.



The calling host dictates the execution strategy:

- **Synchronous evaluation:** The host fulfills the request immediately and invokes `resume(value)`.
- **Asynchronous dispatch:** The host registers the continuation k directly with an OS primitive (such as Linux `io_uring`, macOS `kqueue`, or a libuv event loop). When the kernel signals completion, the host resumes k without blocking threads or altering return types.
- **Preemption and cancellation:** The host can discard k entirely (e.g., during timeouts or aborts), triggering stack unwinding and deterministic resource cleanup.

2. Runtime Platform & Foreign Host Bridging

When running embedded inside an external host (e.g., Java/JVM, JavaScript/Node, WebAssembly/Browser), the host acts as the **ambient platform capability provider**:

- **Host Capability Specialization (given / with):** The compiler checks target platform constraints at build time.

```
// The host environment satisfies GuiContext with native platform hooks
provides GuiContext with WebGLGuiContext()
given Host satisfies Browser & WebGL;
```

- **Foreign Interop Overlays:** Unannotated foreign functions (like JDK classes or npm packages) are assigned capability signatures in the module descriptor (`module.bb`). The host bridges runtime calls between managed algebraic continuations and host runtimes (such as Java Virtual Threads / `ScopedValue` or JS Promise loops).

3. Ambient Context Propagation (Zero-Plumbing)

The calling host injects environmental state into deep call hierarchies without parameter drilling:

- **Context Forwarding:** The host handler wraps computations and attaches request metadata (trace IDs, tenant context, cancellation tokens) to outbound capabilities like `Logger` or `Tracing`.
- **Sandboxed Guest Execution:** A host application can run untrusted guest code or plugins by masking capabilities using type intersections (`performs E & AllowedCaps`), making unauthorized host access impossible.

4. Dynamic Effect Translation & Handshake

The host can translate low-level system events into high-level domain effects (and vice versa):

- A host HTTP server intercepts incoming network sockets and triggers internal business endpoints requiring domain effects (e.g., `OrderService`).
- When the business code performs an ambient capability, a host adapter mixin (extends `Http satisfies GitHub with Mixin()`) translates that domain operation back down into protocol wire calls executed by the host's networking loop.

Summary of Host Responsibilities

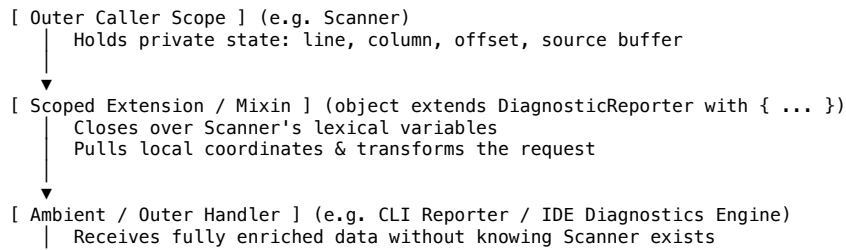
Responsibility	Host Mechanism	Guest/Effect Perspective
I/O & Scheduling	Event loops (<code>io_uring</code> , <code>epoll</code> , <code>Loom</code>)	Synchronous-looking ambient capability
Capability Granting	Module-level grants (<code>grant performs ...</code>)	Statically verified permission
Lifecycle & Cleanup	Continuation scope unwinding	Guaranteed resource finalization
Mocking / Virtualization	Swapping handler instances in <code>handle ... with</code>	Zero code changes across testing and production

User prompt: Yeah, I was not thinking of continuation use cases. More about ability of an effect handler to reach into the calling code context and pull cooperate with the data in

that context. But I guess extension mechanics already offer that capability.

Response: The extension mechanism solves this cleanly.

When you look at cooperation between an effect handler and the calling context without dynamic continuation machinery:



Why Lexical Extensions Outperform Handlers for Context Access

1. **Direct Access to Local State:** A traditional effect handler installed up the stack cannot inspect private fields of an inner class or stack frame without the inner code explicitly passing that state in the effect's argument payload. The object `extends ...` with mixin bridges this boundary inside the lexical scope.
2. **Zero Protocol Bloat:** You don't have to pollute the base capability contract (e.g. `DiagnosticReporter.report(Diagnostic d)`) with scanner-specific or AST-specific parameters (like `line`, `col`, `tokenIndex`, or `sourceFile`).
3. **Unidirectional Simplicity:** The caller enriches the outgoing capability by pulling its own state before delegating upward, avoiding complex bidirectional request/response ping-pong between caller and host handler.

The extension mechanism effectively turns the calling scope into an **active translator** that pulls local context into the ambient effect channel before it ever hits the runtime handler.
